

Chapter 5

Markov Chains

Overview

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Markov Chains

- A Markov process is a random process for which the future (the next step) depends only on the present state; it has no memory of how the present state was reached.
- A Markov chain is a sequence of random values whose probabilities at a time interval depends upon the value of the number at the previous time.
- If the future states of a process are independent of the past and depend only on the present, the process is called a Markov process. A discrete state Markov process is called a Markov chain.
- A Markov Chain is a random process with the property that the next state depends only on the current state.

Markov Chains

- Since the system changes randomly , it is generally impossible to predict the exact state of the system in the future.
- However, the statistical properties of the system's future can be predicted.
- In many applications it is these statistical properties that Markov Chains are important.
- M/M/m queues can be modeled using Markov processes.
- The time spent by the job in such a queue is Markov process and the number of jobs in the queue is a Markov chain.

Markov Chains

- A simple example is the non returning random walk, where the walkers are restricted to not go back to the location just previously visited.
- In this case, each possible position is known as state or condition.
- The controlling factor in a Markov chain is the transition probability, it is a conditional probability for the system to go to a particular new state, given the current state of the system.
- Since the probability of moving from one state to another depends on probability of the preceding state, transition probability is a conditional probability

Markov Chains

- Markov chains is a mathematical tools for statistical modeling in modern applied mathematics, information science.
- Markov chains are used to analyze trends and predict the future. (Weather, stock market, product success, etc.).

5.1 Key Features of Markov Chains

- A sequence of trials of an experiment is a Markov chain if
 - 1) the outcome of each experiment is one of a set of discrete states;
 - 2) the outcome of an experiment depends only on the present state, and not on any past states;
 - 3) the transition probabilities remain constant from one transition to the next.

5.2 Markov Process with Examples

- As we have discussed, we can view a stochastic process as sequence of random variables $\{x_1, x_2, x_3, x_4, x_5, x_6, x_7, \dots\}$.
- Suppose that x_7 depends only on x_6 , x_6 depends only on x_5 , x_5 on x_4 , and so forth.
- In general, if for all i, j, n ,

$$P(x_{n+1} = j \mid x_n = i_n, x_{n-1} = i_{n-1}, \dots, x_0 = i_0) = P(x_{n+1} = j \mid x_n = i_n),$$

then this process is what we call a Markov chain.

5.2 Markov Process with Examples

- The conditional probability above gives us the probability that a process in state i_n at time n moves to i_{n+1} at time $n + 1$. We call this the transition probability for the Markov chain.
- If the transition probability does not depend on the time n , we have a stationary Markov chain, with transition probabilities.

$$P_{ij} = P(X_{n+1} = j | X_n = i)$$

- Now we can write down the whole Markov chain as a matrix P :

$$P = \begin{bmatrix} P_{11} & P_{12} & \dots & P_{1n} \\ P_{21} & P_{22} & \dots & P_{2n} \\ \vdots & \vdots & \ddots & \\ P_{n1} & P_{n2} & & P_{nn} \end{bmatrix}$$

5.2 Markov Process with Examples

- The probability of going from state “i” to state “j” in “n” time steps is:

$$p_{ij}^{(n)} = \Pr(X_n = j \mid X_0 = i)$$

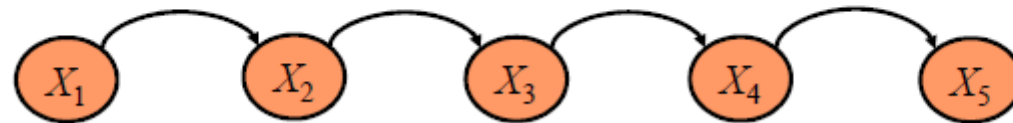
and the single-step transition is

$$p_{ij} = \Pr(X_1 = j \mid X_0 = i).$$

5.2 Markov Process with Examples

- Markov Property: The state of the system at time $t+1$ depends only on the state of the system at time t .

$$\Pr[X_{t+1} = x_{t+1} \mid X_1 \cdots X_t = x_1 \cdots x_t] = \Pr[X_{t+1} = x_{t+1} \mid X_t = x_t]$$



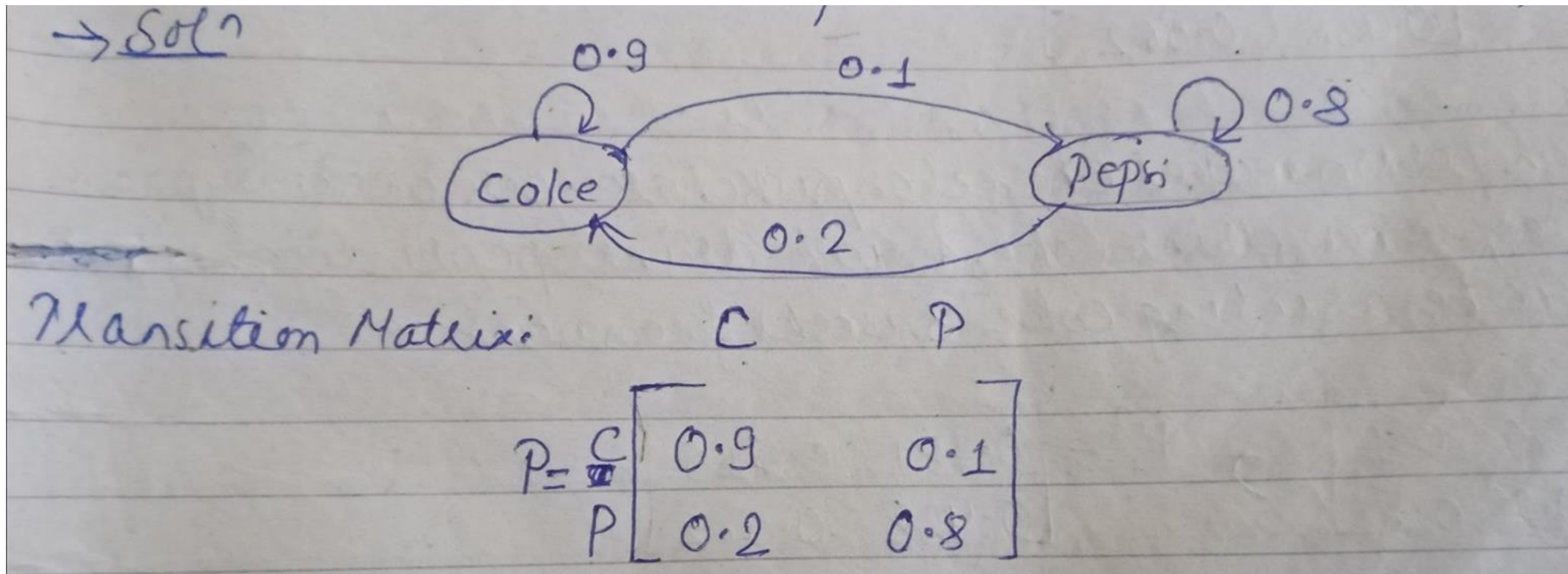
- Stationary Assumption: Transition probabilities are independent of time (t).

$$\Pr[X_{t+1} = b \mid X_t = a] = p_{ab}$$

Markov Process: Example 1

- Coke Vs Pepsi

- Given that a person's last cola purchase was coke, there is a 90% chance that the next cola purchase will also be coke. If a person's last cola purchase was pepsi, there is a 80% chance that the next cola purchase will also be pepsi.



Markov Process: Example 1

- a. Given that a person is currently a pepsi purchaser, what is the probability that he will purchase coke two purchase from now?

$$\begin{aligned}\Pr[\text{Pepsi} \rightarrow ? \rightarrow \text{Coke}] &= \Pr[\text{Pepsi} \rightarrow \text{Coke} \rightarrow \text{Coke}] + \\ &\quad \Pr[\text{Pepsi} \rightarrow \text{Pepsi} \rightarrow \text{Coke}] \\ &= 0.2 \times 0.9 + 0.8 \times 0.2 \\ &= 0.34 \\ &\quad \text{OR} \\ &= \end{aligned}$$

$$P = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$$

$$= \begin{bmatrix} 0.83 & 0.17 \\ \boxed{0.34} & 0.66 \end{bmatrix}$$

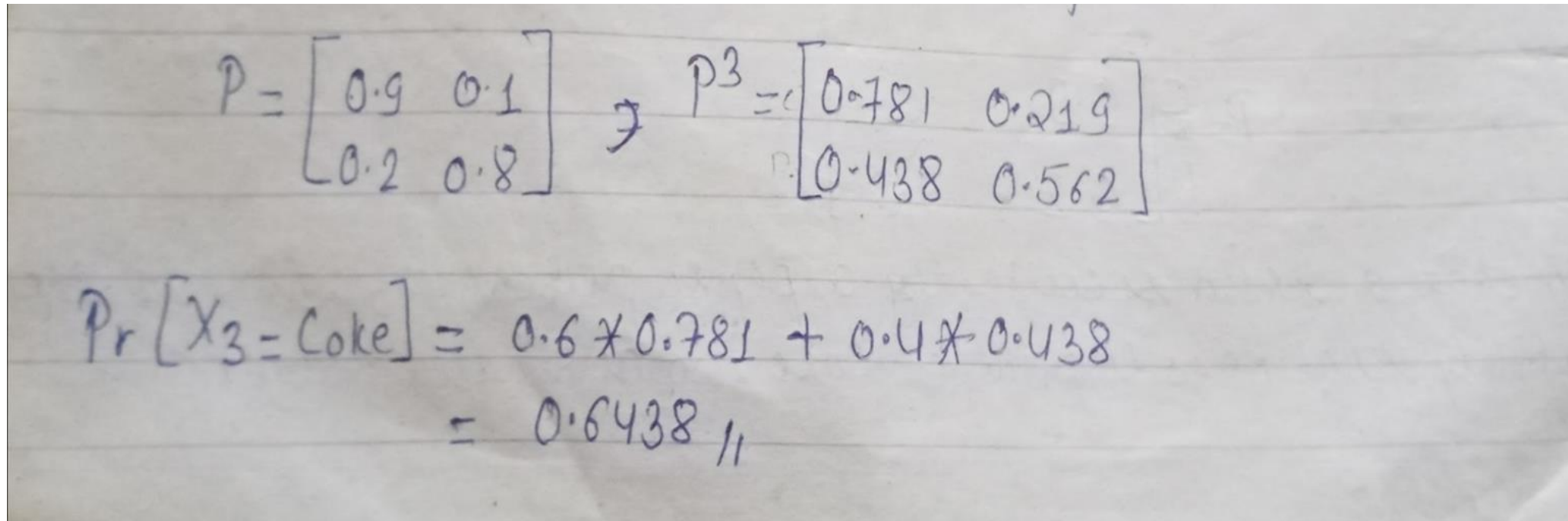
Markov Process: Example 1

b. Given that a person is currently a coke purchaser, what is the probability that he will purchase pepsi after three purchases?

$$P^3 = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \begin{bmatrix} 0.83 & 0.17 \\ 0.34 & 0.66 \end{bmatrix}$$
$$= \begin{bmatrix} 0.781 & 0.219 \\ 0.438 & 0.562 \end{bmatrix}$$

Markov Process: Example 1

c. Assume each person makes one cola purchase per week. Suppose 60% of all people now drink coke, and 40% drink pepsi. What fraction of people will be drinking coke 3 weeks from now?

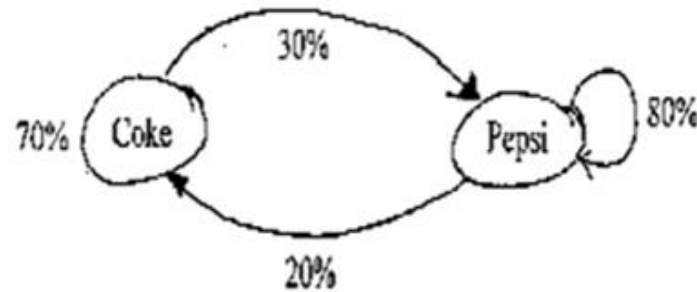


The image shows a handwritten solution on lined paper. It defines a transition matrix P for cola purchases, where the first row represents staying with the current choice and the second row represents switching. $P = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$. It then calculates the 3-week transition matrix $P^3 = \begin{bmatrix} 0.781 & 0.219 \\ 0.438 & 0.562 \end{bmatrix}$. Finally, it computes the probability of drinking Coke after 3 weeks by multiplying the initial state vector $[0.6, 0.4]$ by P^3 , resulting in 0.6438 .

$$P = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \Rightarrow P^3 = \begin{bmatrix} 0.781 & 0.219 \\ 0.438 & 0.562 \end{bmatrix}$$
$$\Pr[X_3 = \text{Coke}] = 0.6 \times 0.781 + 0.4 \times 0.438$$
$$= 0.6438 //$$

Markov Process: Example 2

Example: Given figure shows coke and Pepsi purchaser.



- Given that a person is currently a coke purchaser, what is the probability that he will purchase pepsi after three purchases?
- Assume each person makes one cola purchase per week. Suppose 55% of all people now drink coke, what fraction of people will be drinking coke after 3 weeks?

Markov Process: Example 2

→ Solⁿ
(a)

$$P = \begin{matrix} & \begin{matrix} C & P \end{matrix} \\ \begin{matrix} C \\ P \end{matrix} & \begin{bmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{bmatrix} \end{matrix}$$

Now,

$$P^3 = \begin{bmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{bmatrix} \begin{bmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{bmatrix} \begin{bmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{bmatrix}$$

$$= \begin{bmatrix} 0.55 & 0.45 \\ 0.3 & 0.7 \end{bmatrix} \begin{bmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{bmatrix}$$

$$= \begin{bmatrix} 0.475 & 0.525 \\ 0.35 & 0.65 \end{bmatrix}$$

(b) Solⁿ

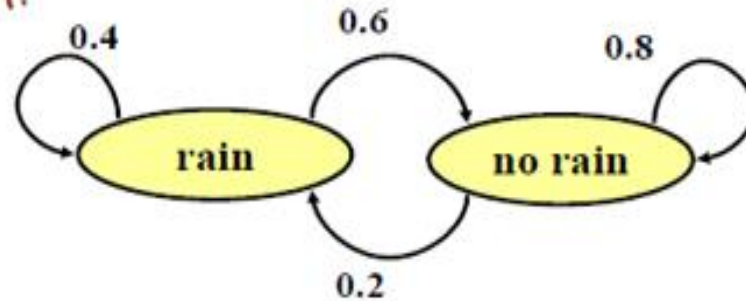
We have; $P = \begin{bmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{bmatrix}$, $P^3 = \begin{bmatrix} 0.475 & 0.525 \\ 0.35 & 0.65 \end{bmatrix}$

$$\begin{aligned} \Pr[X_3 = \text{Coke}] &= 0.55 \times 0.475 + 0.45 \times 0.35 \\ &= 0.26125 + 0.1575 \\ &= 0.41875 \\ &= 41.875\% \end{aligned}$$

Markov Process: Example 3

- Given figure shows the weather condition of rain or no rain.

Stochastic FSM:



- If it is not raining today, find out the probability of having rain a day after tomorrow.

Markov Process: Example 3

Solution,

$$P = \begin{matrix} & \begin{matrix} R & NR \end{matrix} \\ \begin{matrix} R \\ NR \end{matrix} & \begin{bmatrix} 0.4 & 0.6 \\ 0.2 & 0.8 \end{bmatrix} \end{matrix}$$

Now,

$$P^2 = \begin{bmatrix} 0.4 & 0.6 \\ 0.2 & 0.8 \end{bmatrix} \begin{bmatrix} 0.4 & 0.6 \\ 0.2 & 0.8 \end{bmatrix}$$

$$= \begin{bmatrix} 0.28 & 0.72 \\ \textcircled{0.24} & 0.76 \end{bmatrix}$$

$\therefore$ There is 24% chance of raining a day after tomorrow.

What are the key features of Markov Chain? Given that a chance of a MoMo lover to order MoMo in next purchase is 70% and that his next purchase will be Burger is 30% and chance of a Burger lover to order a Burger at the next purchase is 80% and chance that his next purchase will be MoMo is 20%.

a. What is the probability to buy a Burger after three purchase of a current MoMo lover?

b. If 55% of people order MoMo today and 45% people order Burger, what percentage of people will use MoMo after 3 purchase?

Given that a chance of a Honda bike user to buy Honda in next purchase is 70% and that his next purchase will be Yamaha bike is 30% and chance of a Yamaha bike user to buy Yamaha at the next purchase is 50% and chance that his next purchase will be Honda is 50%.

a. What is the probability to buy Yamaha after three purchase of a current Honda bike user?

c. If 50% of user uses Honda today, what percentage of user will use Honda after 3 purchase?

5.3 Applications of Markov Chain

- Used in pattern recognition.
- Used in routing and page rank algorithm.
- Used in weather forecasting.
- Used in stock market prediction.
- Used in forecasting product success.
- Used to study web navigation behavior of internet users.

References

- Jerry Banks, John S. Carson, II Barry L. Nelson , David M. Nicol, P. Shahabudeen “Discrete Event system simulation”